Connor Jaffe

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EDUCATION

Rensselaer Polytechnic Institute, Troy, NY — August 2025 – Expected Graduation: May 2028

Bachelor of Science in **Electrical Engineering** - **3.6 GPA**

- **Relevant Coursework:** Electric Circuits, Computer Systems & Architecture, Embedded Systems, Differential Equations, Calculus

SKILLS

- **Hardware:** Microcontrollers, Soldering, Circuit Design, PID, ADC, I^2C Communication, Op-Amps, Digital Logic, Sequential Circuits
- **Programming Languages:** Python, C, Java, C++, SystemVerilog, HTML/CSS
- **Software & Tools:** MATLAB, LTSpice, Vivado, Siemens NX, Onshape, Simulink, Arduino IDE
- **Languages:** French

PROJECTS

One-Pound Combat Robotics — September 2026 - Current

- **Modeled & constructed** a robot by designing components in **Onshape** & **soldering** electrical systems

Threshold Detector — October 2026

- **Engineered** an **operational amplifier**-based **comparator** circuit to process **dual independent DC inputs**, executing real-time signal evaluation against a **baseline reference voltage threshold**
- **Utilized circuit analysis** techniques and **LTSpice** to stabilize op-amp inputs, ensuring **high-accuracy threshold detection** and optimal switching performance with **minimal signal noise propagation**

Automated Access Control Gate with Dynamic Password Authentication — October 2026

- **Developed** an **embedded security system** utilizing an **Arduino** to orchestrate button-input **password generation**
- **Programmed** an **ultrasonic sensor** in **C++ to** achieve precise **object proximity detection** and trigger an automated **servo motor**
- **Simulated** complete circuit topology within **LTSpice** to validate theoretical calculations using rigorous **circuit analysis** techniques

PID - Remote-Controlled Car — April 2026

- **Designed & programmed** a **microcontroller-powered** RC car **in C**, utilizing a PID control loop to optimize directional stability
- **Integrated analog potentiometers** as control knobs with an **ADC** to translate user input into precise pulse-width modulation signals for motor actuation, speed control, & direction control
- **Optimized PID** control loop algorithms, eliminating steady-state error and minimizing overshoot to maximize vehicular stability

Combinational Digital Logic Circuit — October 2025

- **Engineered a digital logic circuit** to decode & map a binary input to drive a 7-segment display module after **simulating** the circuit in **Vivado** using **SystemVerilog**
- **Constructed** a complete **truth table** & used **Boolean algebra & Karnaugh Maps** to streamline efficiency

EXPERIENCE

Saf-T-Swim, Oceanside, NY — August 2023 - August 2025

Position: Swim Instructor & Lifeguard

- **Designed** & delivered customized lesson plans for **50+ students** across varying age groups, maximizing safety & water comfort
- Directed **25+ graduation ceremonies** celebrating student advancement in **technical stroke mastery**

AWARDS

Scouting America, Oceanside, NY — February 2018 - March 2025

Title: Eagle Scout

- **Managed** a **20+ hour** community infrastructure project, deploying **permanent QR-code** digital markers across a **local church** that routes users to web-based **historical** and descriptive platforms **for individual statues** and **the Stations of the Cross**
- Completed **60+ hours** of community service across the local community
- Appointed **Senior Patrol Leader** and led **30+ scouts** while **orchestrating** weekly meetings